

$$Y_t = e_t + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \phi_3 Y_{t-3}$$

$$e_t \stackrel{\text{b.i.}}{\sim} (0, \sigma_e^2)$$

$$E(Y_t) = 0$$

$$\begin{aligned} \text{Var}(Y_t) &= \text{Var}(e_t) + \phi_1^2 \text{Var}(Y_{t-1}) + \phi_2^2 \text{Var}(Y_{t-2}) + \phi_3^2 \text{Var}(Y_{t-3}) \\ &\quad + 2\text{cov}(e_t, \phi_1 Y_{t-1}) + 2\text{cov}(e_t, \phi_2 Y_{t-2}) + 2\text{cov}(e_t, \phi_3 Y_{t-3}) \\ &\quad + 2\text{cov}(\phi_1 Y_{t-1}, \phi_2 Y_{t-2}) + 2\text{cov}(\phi_1 Y_{t-1}, \phi_3 Y_{t-3}) + 2\text{cov}(\phi_2 Y_{t-2}, \phi_3 Y_{t-3}) \\ \gamma_0 &= \sigma_e^2 + \phi_1^2 \gamma_0 + \phi_2^2 \gamma_0 + \phi_3^2 \gamma_0 + 0 + 0 + 0 + 2\phi_1 \phi_2 \gamma_1 + 2\phi_1 \phi_3 \gamma_2 + 2\phi_2 \phi_3 \gamma_1 \\ &= \sigma_e^2 + (\phi_1^2 + \phi_2^2 + \phi_3^2) \gamma_0 + (2\phi_1 \phi_2 + 2\phi_2 \phi_3) \gamma_1 + 2\phi_1 \phi_3 \gamma_2 \\ &= \sigma_e^2 + (\phi_1^2 + \phi_2^2 + \phi_3^2) \gamma_0 + 2\phi_2 (\phi_1 + \phi_3) \gamma_1 + 2\phi_1 \phi_3 \gamma_2 \end{aligned}$$

$$\begin{aligned} \gamma_1 &= \text{cov}(Y_t, Y_{t-1}) \\ &= \text{cov}(e_t + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \phi_3 Y_{t-3}, Y_{t-1}) \\ &= \phi_1 \gamma_0 + \phi_2 \gamma_1 + \phi_3 \gamma_2 \end{aligned}$$

$$\begin{aligned} \gamma_2 &= \text{cov}(Y_t, Y_{t-2}) \\ &= \text{cov}(e_t + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \phi_3 Y_{t-3}, Y_{t-2}) \\ &= \phi_1 \gamma_1 + \phi_2 \gamma_0 + \phi_3 \gamma_1 \end{aligned}$$

$$\begin{aligned} \gamma_3 &= \text{cov}(Y_t, Y_{t-3}) \\ &= \text{cov}(e_t + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \phi_3 Y_{t-3}, Y_{t-3}) \\ &= \phi_1 \gamma_2 + \phi_2 \gamma_1 + \phi_3 \gamma_0 \end{aligned}$$

$$\gamma_k = \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2} + \phi_3 \gamma_{k-3} //$$

➤ Mencari γ_0 teoritis

Substitusi γ_2 ke γ_1

$$\gamma_1 = \phi_1 \gamma_0 + \phi_2 \gamma_1 + \phi_3 (\phi_1 \gamma_1 + \phi_2 \gamma_0 + \phi_3 \gamma_1)$$

$$\gamma_1 = \phi_1 \gamma_0 + \phi_2 \phi_3 \gamma_0 + \phi_2 \gamma_1 + \phi_1 \phi_3 \gamma_1 + \phi_3^2 \gamma_1$$

$$(1 - (\phi_2 + \phi_1 \phi_3 + \phi_3^2)) \gamma_1 = (\phi_1 + \phi_2 \phi_3) \gamma_0$$

$$\gamma_1 = \frac{(\phi_1 + \phi_2 \phi_3) \gamma_0}{(1 - (\phi_2 + \phi_1 \phi_3 + \phi_3^2))}$$

$$\begin{aligned} \gamma_2 &= \frac{\phi_1 (\phi_1 + \phi_2 \phi_3) \gamma_0}{(1 - (\phi_2 + \phi_1 \phi_3 + \phi_3^2))} + \phi_2 \gamma_0 + \frac{\phi_3 (\phi_1 + \phi_2 \phi_3) \gamma_0}{(1 - (\phi_2 + \phi_1 \phi_3 + \phi_3^2))} \\ &= \left(\frac{\phi_2 + (\phi_1 + \phi_3) (\phi_1 + \phi_2 \phi_3)}{(1 - (\phi_2 + \phi_1 \phi_3 + \phi_3^2))} \right) \gamma_0 \end{aligned}$$

$$= \left(\frac{\phi_2 + (\phi_1 + \phi_3)(\phi_1 + \phi_2\phi_3)}{1 - (\phi_2 + \phi_1\phi_3 + \phi_3^2)} \right) \gamma_0$$

$$\gamma_0 = \sigma e^2 + (\phi_1^2 + \phi_2^2 + \phi_3^2) \gamma_0 + 2\phi_2(\phi_1 + \phi_3) \gamma_1 + 2\phi_1\phi_3 \gamma_2$$

$$= \sigma e^2 + (\phi_1^2 + \phi_2^2 + \phi_3^2) \gamma_0 + \frac{2\phi_2(\phi_1 + \phi_3)(\phi_1 + \phi_2\phi_3)}{1 - (\phi_2 + \phi_1\phi_3 + \phi_3^2)} \gamma_0 + 2\phi_1\phi_3 \left(\frac{\phi_2 + (\phi_1 + \phi_3)(\phi_1 + \phi_2\phi_3)}{1 - (\phi_2 + \phi_1\phi_3 + \phi_3^2)} \right) \gamma_0$$

$$= \sigma e^2 + (\phi_1^2 + \phi_2^2 + \phi_3^2) \gamma_0 + \frac{2\phi_1\phi_2\phi_3 + 2(\phi_2 + \phi_1\phi_3)(\phi_1 + \phi_3)(\phi_1 + \phi_2\phi_3)}{1 - (\phi_2 + \phi_1\phi_3 + \phi_3^2)} \gamma_0$$

$$= \sigma e^2 + \left(\frac{(\phi_1^2 + \phi_2^2 + \phi_3^2) + 2(\phi_1\phi_2\phi_3 + (\phi_2 + \phi_1\phi_3)(\phi_1 + \phi_3)(\phi_1 + \phi_2\phi_3))}{1 - (\phi_2 + \phi_1\phi_3 + \phi_3^2)} \right) \gamma_0$$

$$\frac{(1 - (\phi_2 + \phi_1\phi_3 + \phi_3^2)) - (1 - (\phi_2 + \phi_1\phi_3 + \phi_3^2))(\phi_1^2 + \phi_2^2 + \phi_3^2) - 2(\phi_1\phi_2\phi_3 + (\phi_2 + \phi_1\phi_3)(\phi_1 + \phi_3)(\phi_1 + \phi_2\phi_3))}{-11} \gamma_0 = \sigma e^2$$

$$\gamma_0 = \frac{(1 - (\phi_2 + \phi_1\phi_3 + \phi_3^2)) - 2(\phi_1\phi_2\phi_3 + (\phi_2 + \phi_1\phi_3)(\phi_1 + \phi_3)(\phi_1 + \phi_2\phi_3))}{-11} = \sigma e^2$$

$$\gamma_0 = \frac{\sigma e^2 (1 - (\phi_2 + \phi_1\phi_3 + \phi_3^2))}{(1 - (\phi_2 + \phi_1\phi_3 + \phi_3^2)) (1 - (\phi_1^2 + \phi_2^2 + \phi_3^2)) - 2(\phi_1\phi_2\phi_3 + (\phi_2 + \phi_1\phi_3)(\phi_1 + \phi_3)(\phi_1 + \phi_2\phi_3))}$$

⇒ Mencari korelasi

$$\rho_k = \frac{\gamma_k}{\gamma_0}$$

⇒ diperoleh sebelumnya bahwa

$$\gamma_k = \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2} + \phi_3 \gamma_{k-3}$$

$$\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2} + \phi_3 \rho_{k-3} \text{ untuk } k = 0, 1, 2, 3, \dots$$

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